Verification Plan

SyoSil, 2026

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Introduction

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Introduction

- A verification plan can be many things
  - Typically a mapping of requirements to verification targets

- For a **directed test** approach
  - Mapping of requirements to tests
  - Result is a list of tests to be implemented

- For **CRV**
  - Mapping of requirements to **coverage** and **checkers**
  - Result is **NOT** a list of tests to be implemented
    - CRV VPs can include named tests and targeted sequences.
  - In general, whatever tests closes the coverage model with all checkers enabled will do the job

Both approaches need requirement mapping and checking.

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Purpose of a Verification Plan

A mapping of design requirements to verification evidence

Design Requirement	Which specified behavior does the entry address?
Verification Criteria	Which checker determines whether the behaviour is correct?
Coverage	Which observations show that we exercised the behaviour?

Review the mapping before implementing the testbench.

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Tools?

- To handle requirement mapping some tools are available
  - DOORS: Requirement management
  - Excel: Poor mans solution
  - Proprietary format (TXT, JSON, …)
    - Needs some proprietary scripting as well
  - EDA Tools
    - VManager (Cadence)
    - HVP – Hierarchical Verification Plan (Synopsys)
    - TestPlan Author (Siemens-EDA)

- We will use Excel (Or similar)

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Verification Plan

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Verification Plan (VP)

- A verification plan (VP) is **requirement mapping** to coverage, checkers and other testbench components
  - Will ensure that the requirements to the RTL design are met

- VPs in real life can be very long and complex
  - We have chosen to use an easy/simple VP format
    - simple Verification Plan (sVP) format

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simple Verification Plan Format (sVP)

ID	Design Requirement	Coverage	Verification Criteria	Priority	Type	Comment
<int>	Requirement text or ID	Covergroup, Coverpoint, Assertion coverage, Cover property, structural coverage or manual inspection	Checker, scoreboard, assertion or reference model, manual inspection	High Medium Low	Design Protocol	Scenario, sampling rule or note

- **ID:** Each entry has a unique ID
- **Design Requirement:** Either a requirement text or reference a design requirement
- **Coverage:** Can be a reference to a coverage type or to the actual implementation
  - WRT manual inspection. This is the last resort if automatic coverage cannot be implemented. We should try not to use that.
- **Verification Criteria:** Explains how correctness is checked.
  - Same WRT manual inspection as for coverage

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simple Verification Plan Format (sVP)

ID	Design Requirement	Coverage	Verification Criteria	Priority	Type	Comment
<int>	Requirement text or ID	Covergroup, Coverpoint, Assertion coverage, Cover property, structural coverage or manual inspection	Checker, scoreboard, assertion or reference model, manual inspection	High Medium Low	Design Protocol Other?	Scenario, sampling rule or note

- **Priority:** Speaks for itself
- **Type:** What is this related to?
  - Typically, the Design, Protocols involved but might also be other relevant types
- **Comment:** can describe the scenario or sampling condition or other comments

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Example
sSDT Protocol sVP

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sSDT Protocol Requirements

All requirements apply at the sampled clock edge
See: exercise_general_info.pdf

PR1 rst == 1: valid cannot be 1

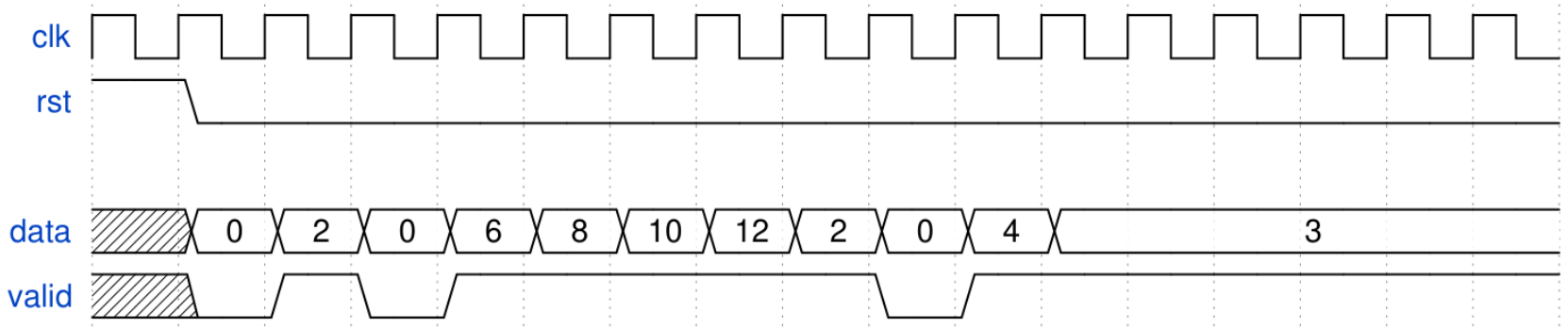
So, it can be X/Z and 0

PR2 valid == 1: data must have a valid value

rst == 0 implied

Valid data means no X/Z here. Zero is a legal value.

PR3 valid == 0: data must be 0



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Example: sSDT verification plan

ID	Design Requirement	Coverage	Verification Criteria	Priority	Type	Comment
0	PR1	Reset + release; entry: idle, active	Reset checker: rst==1: valid==!1	High	Protocol	Check at each rising edge
1	PR2	Data: 0, [1:MAX-1], MAX	Data checker: (rst==0 &&) valid==1: no X/Z	High	Protocol	Sample data bins when valid==1
2	PR3	Idle seen; active-to-idle	Idle checker: valid==0: data==0	High	Protocol	Also applies during reset

Requirement text: PR1–PR3 on the preceding slide. MAX is the largest data value.

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sSDT Coverage Related to Tests

PR1: reset – Covered by standard random test

Assertion and release. Reset entry from idle and active traffic.

PR2: valid data – Covered by standard random test

Data bins 0, MAX and other values

PR3: idle – Covered by standard random test

Idle observed. Active-to-idle transition and continued idle.

Legal rst/valid combinations: 00, 01, 10.
Rst==1 and valid==1 is a violation, never a closure target.

One single random test should be able to cover PR1-PR3

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sSDT Checkers – Pseudo Code

Checker logic after clocked sampling of rst, valid and data signals

```
if rst_high:
   check(valid_is_zero, "PR1")
if valid_is_one:
   check(data_known, "PR2")
if valid_is_zero:
   check(data_known and data_value == 0, "PR3")
```

Report each violation with its requirement ID.

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Example of Coverage and Checkers Working Together

Observed at a clock edge: valid = 0, data = 5

Coverage result	The idle scenario occurred. Data only sampled when valid==1 – **NO FALSE POSITIVE** ☺
PR3 checker result	Failure: idle data must be zero.

- **A coverage hit and a protocol failure can occur together if coverage is not guarded correctly.**
- **If coverage not guarded then important that flow excludes coverage from failing tests from coverage report**

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Coverage Closure Following the sVP

- All required coverage bins have been observed
- PR1, PR2 and PR3 checks are active and passing
- Results can be reproduced in a regression
  - Store versions, configurations, seed in the verification flow
- Handle unhit bins (or other unclosed coverage)
  - May need better stimulus
  - A coverage fix
  - An applicability review.
  - Exclusions – **last resort!**
- Review of final coverage report

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Exercise: E12 - SAT verification plan

Create the SAT plan from requirements DR/01–DR/06.
See: exercise_general_info.pdf

Use the same seven columns. Reuse the sSDT plan for protocol compliance.
Record any unclear requirement as a question for review.

ID	Design Requirement	Coverage	Verification Criteria	Priority	Type	Comment
	DR...					
	DR...					

Add as many entries as the requirements need.

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